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Application of Machine Learning Methods to the Detection of Early Warning Signals of Atlantic Meridional Overturning Circulation Weakening

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Abstract

The Atlantic Meridional Overturning Circulation (AMOC) is a vital component of the Earth's climate system but recent evidence suggests it may be weakening. Detection of early warning signals of a weakening AMOC remains a scientific challenge. This project explores the application of machine learning methods to identify potential early warning signals using ocean and atmospheric reanalysis datasets. Statistical analysis and machine learning methods are applied to historical climate variables to identify whether changes in predictive performance can evidence evolving AMOC behaviour.

Contents

1 Introduction and Background

1.1 The Atlantic Meridional Overturning Circulation

The Atlantic Meridional Overturning Circulation (AMOC) is one of the most important components of the Earth's climate system. It consists of a series of ocean currents that transport warm, saline surface waters northward through the Atlantic Ocean, while returning colder, denser waters southward at depth. This circulation plays a vital role in the regulation of regional and global climate conditions via the redistribution of enormous quantities of heat, freshwater and carbon around the globe. Approximately 1 petawatt (10^{15}Js^{-1}) of heat is transported northward across the Atlantic by the AMOC making it a major contributor to the relatively mild climate impacts of a slowdown [1, 2].

The AMOC is a subsection of the global thermohaline circulation, often referred to as the "global ocean conveyor belt" [1], which is driven by differences in seawater density. This seawater density is controlled primarily by temperature and salinity, with cold, salty water being denser than warm or fresh water. Surface water originating in the tropics is warmed by incident solar radiation and transported northward by the AMOC and Gulf Stream. Throughout it's journey, this surface water releases heat to the atmosphere, particularly over the North Atlantic, causing it to cool. Simultaneously, evaporation increases its salinity. These processes work in combination to increase water density sufficiently for deep convection to occur in the Labrador Sea, Irminger Sea and Nordic Seas, where dense water sinks to form the North Atlantic Deep Water (NADW) [1, 2]. This deep water subsequently flows southward before eventually returning to the surface elsewhere through the upwelling process, completing the overturning circulation.

The AMOC influences the global climate on timescales ranging from decades to millennia via heat transport, ocean ventilation and carbon storage. By transporting warm water northwards, it moderates winter temperatures across western Europe and affects atmospheric circulation patterns, storm tracks and precipitation. Beyond temperature regulation the circulation redistributes nutrients and oxygen, contributing to marine ecosystems, while also acting as a major component of the global carbon cycle through the storage and transport of dissolved carbon. As a result, changes in AMOC strength can have widespread implications [1, 2].

Paleoclimatic evidence demonstrates that the AMOC has not remained constant throughout Earth's history. During the last glacial period, multiple abrupt climate transitions, known as Dansgaard-Oeschger evets, were associated with rapid shifts between strong and weak AMOC modes. Geological records indicate that these transitions occurred over relatively short timescales and were accompanied by changes in temperature exceeding 5–10°C across Greenland, alongside changes in precipitation patterns, sea ice and climate on a global scale. These historical transitions demonstrate that the AMOC can possess multiple stable states and, when critical thresholds are exceeded, undergoing abrupt and rapid restructuring [1, 5].

Future climate projections consistently suggest that anthropogenic climate change will weaken the AMOC. Increasing atmospheric temperatures lead to increasing freshwater forcing of the North Atlantic via enhanced precipitation and accelerated melting of the Greenland Ice Sheet. This freshwater introductions reduces surface salinity and therefore density, inhibiting sinking and the formation of deep water and weakening the overturning circulation. Although most current climate models predict a gradual reduction of AMOC strength in the twenty-first century rather than a complete shutdown, uncertainty remains regarding bot the magnitude of this weakening and the possibility of abrupt transitions (a result of known biases in many climate models) [2–4].

Given the important role the AMOC plays in thermal regulation of the Northern Hemisphere climate and past evidence that it may exhibit non-linear behaviour, understanding whether the AMOC is approaching a critical transition has become one of the most important questions in recent climate study. The detection of reliable early warning signals of future weakening or collapse therefore represents a significant scientific challenge, with major implications regarding climate prediction and mitigation [5, 6].

1.2 Evidence for AMOC Weakening

Since the late twentieth century, increasing evidence suggests that the AMOC is weakening in response to anthropogenic climate change. Despite direct AMOC observations being limited, a combination of instrumental measurements, ocean reanalysis datasets, paleoclimate reconstructions and climate model simulations all indicate a decline in AMOC strength. Significant uncertainty remains regarding the timing and magnitude of this weakening. However, it is broadly agreed upon that continued global warming is expected to reduce AMOC strength throughout the twenty-first century [2, 3, 7].

A major difficulty regarding the assessment of long-term AMOC behaviour is the lack of continuous direct observations. Comprehensive, direct measurements of the AMOC have only been available since 2004 through the RAPID-MOCHA array (located at approximately 26.5°N in the Atlantic Ocean). While observations from this array have provided an unprecedented view of AMOC variability, the relatively short duration of the record results in inter-annual and decadal variability dominating the data, making long-term climate trends difficult to distinguish. Nonetheless, RAPID observations have still demonstrated substantial variability in AMOC strength and revealed periods of anomalously weak circulation [6, 7].

In order to overcome these limitations, researchers have developed methods of reconstructing historical AMOC strength indirectly using observable "fingerprints". For example, AMOC strength can be derived from characteristic patterns in North Atlantic sea surface temperature (SST). Due to the AMOC transporting large quantities of heat northwards, a weakening of the circulation leads to decreased heat transport to the subpolar North Atlantic, resulting in a distinctive pattern consisting of cooling south of Greenland and warming along the Gulf Stream region of the western Atlantic. Caesar et al. demonstrated that this fingerprint is present in observational SST datasets and closely resembles that produced by climate simulations of the AMOC weakening. Working on this relationship, it is estimated that the AMOC has weakened by approximately 3 ± 1 Sverdrups, equivalent to roughly 15% since the mid-twentieth century. Their reconstructed index also indicates that recent decades contain the weakest AMOC state of the observational record [7].

Oceanographic and paleoclimate proxy records also offer independent evidence of a weakening AMOC. Analyses of marine sediments, coral records and isotopic tracers suggest that present AMOC strength is unusually weak when compared with conditions during the last millennium. Proxy reconstructions indicate that, despite the AMOC naturally experiencing variability on decadal to centennial timescales, recent changes appear consistent with external forcing associated with anthropogenic climate change rather than internal variability alone. In addition, geological evidence demonstrates the AMOC's ability to shift between strong and weak modes during previous glacial periods, resulting in abrupt climatic transitions over short timescales. This historical evidence confirms that large changes in AMOC strength are possible within the Earth system [1, 5].

Climate model simulations provide further evidence of AMOC weakening as a response to anthropogenic climate change. While the magnitude of projected weakening varies considerably between models, almost all CMIP5 and CMIP6 simulations predict a reduction in AMOC strength under increasing greenhouse gas concentrations during the twenty-first century. The Intergovernmental Panel on Climate Change (IPCC) Fifth Assessment Report concluded that the AMOC is *very likely* to weaken over the coming century under all major emissions scenarios, although a complete collapse before 2100 was assessed as *very unlikely* [3]. Despite this broad agreement, AMOC stability remains a contentious topic. More recent studies have suggested that many climate models underestimate the sensitivity of the AMOC as a result of biases in the representation of freshwater transport, ocean mixing, and deepwater formation and correcting these biases has been shown to fundamentally alter the simulated response to increasing CO_2 concentrations. Liu et al. demonstrated that, rather than exhibiting a gradual weakening, the corrected model eventually experienced a complete AMOC collapse, highlighting that model biases may underestimate true AMOC sensitivity [4, 8].

Similarly, North Atlantic convection studies have implied that many climate models underestimate the likelihood of abrupt regional cooling due to reduced deep water formation. Climate models capable of more realistic North Atlantic stratification reproduction were significantly ore likely to experience abrupt cooling associated with weak convection with results implying that current models, less suited to ocean stratification and deep convection simulation, may underestimate the probability of rapid changes in the North Atlantic [8].

Owing to the AMOC's far reaching affects, the implications of continued AMOC weakening extend far beyond the North Atlantic. Simulations indicate that, beyond a cooling across the North Atlantic, a substantial reduction in AMOC strength would lead to changes in atmospheric circulation, southward displacement of tropical rainfall belts, altered storm tracks and significant impacts on European temperature and precipitation. Further consequences include regional sea level rise along the eastern coast of North America, reduced marine life productivity and potential interactions with other tipping elements such as the Greenland Ice Sheet and Amazon rainforest [2, 9].

Although the exact future evolution of the AMOC remains uncertain, the collective evidence from observations, reanalysis', proxy records ad climate models strongly suggest a weakening circulation, with recent studies indicating that the present day AMOC may be exhibiting characteristics associated with declining stability and increasing susceptibility to abrupt transition. Detecting reliable early warning signals capable of identifying these changes before a critical threshold is reached therefore represents a vital challenge in climate science and one that forms the central motivation for this study.

1.3 Tipping Points

1.4 Early Warning Signals

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3 Model Performance

4 Conclusion

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References

[1] Rahmstorf, S. (2002). Ocean circulation and climate during the past 120,000 years. *Nature*, 419(6903), 207–214.

[2] Jackson, L. C., Kahana, R., Graham, T., Ringer, M. A., Woollings, T., Mecking, J. V., & Wood, R. A. (2015). Global and European climate impacts of a slowdown of the AMOC in a high resolution GCM. *Climate Dynamics*, 45, 3299–3316.

[3] Collins, M., Knutti, R., Arblaster, J., Dufresne, J.-L., Fichefet, T., Friedlingstein, P., et al. (2013). Long-term Climate Change: Projections, Commitments and Irreversibility. In: *Climate Change 2013: The Physical Science Basis*. Cambridge University Press.

[4] Liu, W., Xie, S.-P., Liu, Z., & Zhu, J. (2017). Overlooked possibility of a collapsed Atlantic Meridional Overturning Circulation in warming climate. *Science Advances*, 3(1), e1601666.

[5] Boers, N. (2021). Observation-based early-warning signals for a collapse of the Atlantic Meridional Overturning Circulation. *Nature Climate Change*, 11, 680–688.

[6] van Westen, R. M., Kliphuis, M., & Dijkstra, H. A. (2024). Physics-based early warning signal shows that AMOC is on tipping course. *Science Advances*, 10(6), eadk1189.

[7] Caesar, L., Rahmstorf, S., Robinson, A., Feulner, G., & Saba, V. (2018). Observed fingerprint of a weakening Atlantic Ocean overturning circulation. *Nature*, 556, 191–196.

[8] Sgubin, G., Swingedouw, D., Drijfhout, S., Mary, Y., & Bennabi, A. (2017). Abrupt cooling over the North Atlantic in modern climate models. *Nature Communications*, 8, 14375.

[9] Drijfhout, S. (2015). Competition between global warming and an abrupt collapse of the AMOC in Earth's energy imbalance. *Scientific Reports*, 5, 14877.

[10] Lenton, T. M., Held, H., Kriegler, E., Hall, J. W., Lucht, W., Rahmstorf, S., & Schellnhuber, H. J. (2008). Tipping elements in the Earth's climate system. *Proceedings of the National Academy of Sciences*, 105(6), 1786–1793.

[11] Scheffer, M., Bascompte, J., Brock, W. A., Brovkin, V., Carpenter, S. R., Dakos, V., Held, H., van Nes, E. H., Rietkerk, M., & Sugihara, G. (2009). Early-warning signals for critical transitions. *Nature*, 461, 53–59.